

## UNIT-1

## LOGICS AND PROOFS

4/10/24

- 1) CONJUNCTION (AND)  $\wedge$
- 2) DISJUNCTION (OR)  $\vee$
- 3) IMPLICATION  $\rightarrow$

• if  $p$  then  $q$ ,  $p$  is sufficient for  $q$   
and  $q$  is necessary for  $p$

- 4) BICONDITIONAL  $\leftrightarrow$

- $p$  iff (if and only if)  $q$ .
- for any unary statements, we have negation that is represented by '!' or 'n'.

### → STATEMENT OR PROPOSITION

- \* Any declarative sentence that evaluates to either true or false, but not both is called a statement.
- \* The statements are represented using lower case letters.

### → DETERMINE IF EACH OF THE FOLLOWING IS A STATEMENT OR NOT

- 1) in 1990, George Bush was the president of the US.  
STATEMENT

- 2) What a beautiful morning.  
NOT A STATEMENT

- it might not be the same for everyone.

3)  $n+3$  is a positive integer.

NOT A STATEMENT

- The value of ' $n$ ' is unknown. It could be a fraction, polynomial, integer etc.

4) 60 is an even number

STATEMENT

- it is a statement, but it is false.

5) If only every morning would be as sunny and clear as today

NOT A STATEMENT

6) What time is it?

NOT A STATEMENT

→ WRITE THE FOLLOWING IN SYMBOLIC FORM:

i)  $p$ : I finished writing my computer program before lunch.

$q$ : I shall play tennis this afternoon.

$r$ : The sun is shining.

$s$ : The humidity is low.

Q. If the sun is shining then I shall play tennis in the afternoon.

$r \rightarrow q$

Q. Low humidity and sunshine are sufficient for me,



\_/\_/\_

to play tennis this afternoon.

$$(r \wedge s) \rightarrow q$$

- Q. Finishing writing my computer program before lunch is necessary for playing tennis in the afternoon.

$$q \rightarrow p$$

• as  $q$  is necessary to happen.

- 2) S: Mary goes out for a walk.

T: The moon is out.

u: It is snowing.

Q.  $(t \wedge \neg u) \rightarrow s$

if the moon is out and not snowing, then Mary goes out for a walk.

Q.  $t \rightarrow (u \rightarrow s)$

if the moon is out then not snowing is sufficient for Mary to <sup>out</sup> go for a walk

Q.  $\neg(s \leftrightarrow (u \vee t))$

\*~~Mary~~ It is not true that Mary goes out for a walk iff it's snowing or the moon is out.

Q.  $s \leftrightarrow t$

Mary goes out for a walk iff (if and only if) the moon is out.

Q.  $(u \wedge \neg t) \rightarrow \neg s$

\_/\_/\_

If it is snowing and the moon is out then Mary does not go out for a walk

imp

Q.

$$t \rightarrow (\sim u \rightarrow s)$$

- \* If the moon is out then not snowing is sufficient for Mary to go out for a walk.
- \* If it is not snowing then Mary goes out for a walk which is necessary for the moon to be out.
- \* If the moon is out, then if it is not snowing then Mary goes out for a walk.

Q.

Mary goes out for a walk iff the moon is out.  
 $s \leftrightarrow t$

Q.

It is snowing but Mary still goes out for a walk.  
 $u \wedge s$        $\hookrightarrow$  But  $\rightarrow$  And

$\rightarrow$

TRUTH TABLES FOR:

| P | q | $P \rightarrow q$ | $P \leftrightarrow q$ |
|---|---|-------------------|-----------------------|
| T | T | T                 | T                     |
| F | F | T                 | T                     |
| T | F | F                 | F                     |
| F | T | F $\rightarrow$ * | F                     |

Q.

Let  $p$  and  $q$  be primitive statements for which  $p$  implies  $q$  is false, then determine the truth values for the following:



$$i) p \wedge q \rightarrow F$$

$$ii) \sim p \vee q \rightarrow F$$

$$iii) q \rightarrow p \rightarrow T$$

$$iv) \sim q \rightarrow \sim p \rightarrow F$$

Q. Determine truth values of each of the following implications:

$$i) \text{ if } 3+4=12 \text{ then } 3+2=6$$

T

$$ii) \text{ If } 3+3=6 \text{ then } 3+6=9$$

T

$$iii) \text{ If } 3+3=6 \text{ then } 3+4=9$$

F

Q. Construct the truth table for the following statements where  $p, q, r$  denote primitive statements.

$$i) \sim (p \vee \sim q) \rightarrow \sim p$$

| p | q |
|---|---|
| T | T |

|   |   |
|---|---|
| F | F |
|---|---|

|   |   |
|---|---|
| T | F |
|---|---|

|   |   |
|---|---|
| F | T |
|---|---|

All true =  
 tautology.  
 All false =  
 contradiction

$$ii) p \rightarrow (q \rightarrow r)$$

| p | q | r |
|---|---|---|
| T | T | T |
| F | F | F |

→ WITH HELP OF TRUTH TABLE VERIFY THE FOLLOWING

Q.  $p \rightarrow q \Leftrightarrow \sim p \vee q$   
 $\searrow$  logic equivalence

| $P$ | $q$ | $P \rightarrow q$ | $\neg P$ | $\neg P \vee q$ |
|-----|-----|-------------------|----------|-----------------|
| 0   | 0   | T                 | 1        | 1               |
| 0   | 1   | T                 | 1        | 1               |
| 1   | 0   | F                 | 0        | 0               |
| 1   | 1   | T                 | 0        | 1               |

$p \rightarrow q$  and  $\neg p \vee q$  same outputs.

Since, the truth table for both the LHS and RHS  
i.e.  $P \rightarrow q$  and  $\neg p \vee q$  are same. They are logically  
equivalent.

a.  $(p \rightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$

| $P$ | $q$ | $P \rightarrow q$ | $q \rightarrow P$ | $(P \rightarrow q) \wedge (q \rightarrow P)$ |
|-----|-----|-------------------|-------------------|----------------------------------------------|
| 0   | 0   | 1                 | 1                 | 1                                            |
| 0   | 1   | 1                 | 0                 | 0                                            |
| 1   | 0   | 0                 | 1                 | 0                                            |
| 1   | 1   | 1                 | 1                 | 1                                            |

No, they are not logically equivalent.

Q.  $(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$

| $\neg p$ | $p$ | $\neg q$ | $q$ | $(p \rightarrow q) \wedge (q \rightarrow p)$ | $p \leftrightarrow q$ |
|----------|-----|----------|-----|----------------------------------------------|-----------------------|
| 1        | 0   | 1        | 0   | 1                                            | 1                     |
| 1        | 0   | 0        | 1   | 0                                            | 0                     |
| 0        | 1   | 1        | 0   | 0                                            | 0                     |
| 0        | 1   | 0        | 1   | 1                                            | 1                     |



$$(p \rightarrow q) \wedge (q \rightarrow p) \quad (\neg p \vee q) \wedge (\neg q \vee p)$$

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 |

yes, they are logically equivalent.

Remember them.

NOTE:

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

$$\Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$$

They are logically equivalent.

## → L A W S    O F    L O G I C

1) Law of double negativity:

$$\neg \neg p \Leftrightarrow p$$

2) De Morgan's Law:

$$\neg (p \vee q) \Leftrightarrow \neg p \wedge \neg q$$

$$\neg (p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

3) Associative Law:

$$p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$$

$$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$$

4) Commutative Law:

$$p \vee q \Leftrightarrow q \vee p$$

$$p \wedge q \Leftrightarrow q \wedge p$$

5) Distributive Law:

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

6) Idempotent Law:

$$p \vee p \Leftrightarrow p$$

$$p \wedge p \Leftrightarrow p$$

7) Identity Law:

$$p \vee F \Leftrightarrow p$$

$$p \wedge T \Leftrightarrow p$$

8) Inverse Law:

$$p \vee \neg p \Leftrightarrow T$$

$$p \wedge \neg p \Leftrightarrow F$$

9) Domination Law:

$$p \vee T \Leftrightarrow T$$

$$p \wedge F \Leftrightarrow F$$

10) Absorption Law:

$$p \vee (p \wedge q) \Leftrightarrow p$$

$$p \wedge (p \vee q) \Leftrightarrow p$$



Q. Negate and simplify  $(p \vee q) \rightarrow r$ .

$$(p \vee q) \rightarrow r$$
$$\sim [\underbrace{(p \vee q)}_p \rightarrow \underbrace{r}_q]$$

Reason:  $p \rightarrow q \Leftrightarrow \sim p \vee q$

$$\sim [\sim (p \vee q) \vee r]$$

$\sim \sim (p \vee q) \wedge \sim r \rightarrow$  Demorgan's Law

$(p \vee q) \wedge \sim r \rightarrow$  Double negation.

$$(p \wedge \sim r) \vee (q \wedge \sim r) //$$

Q. Simplify the following statements:

1)  $(p \vee q) \wedge \sim (\sim p \wedge q)$

Reason

$$(p \vee q) \wedge \sim \sim p \vee \sim q$$

De Morgan's.

$$(p \vee q) \wedge (p \vee \sim q)$$

Double Negation.

$$p \vee (q \wedge \sim q)$$

Distributive

$$p \vee F$$

Inverse

$$p$$

Identity

2)  $\sim (p \vee (\sim p \wedge q))$

Reason

$$\sim (p \vee \sim p) \wedge \sim (p \vee q)$$

Distributive.

$$\sim (T \wedge (p \vee q))$$

$$p \vee \sim p = T$$

$$\sim (p \vee q)$$

$$T \wedge p = p$$

$$\sim p \wedge \sim q$$

Demorgan's Law.

$\rightarrow$

$$\begin{aligned}
 3) & \quad \neg(\neg((p \vee q) \wedge r) \vee \neg q) \\
 & \quad \neg\neg((p \vee q) \wedge r) \wedge \neg\neg q \\
 & \quad ((p \vee q) \wedge r) \wedge q \\
 & \quad (p \vee q) \wedge (r \wedge q) \\
 & \quad (p \vee q) \wedge (q \wedge r) \\
 & \quad ((p \vee q) \wedge q) \wedge r \\
 & \quad (q \wedge r)
 \end{aligned}$$

Reason

De Morgan's

Double negation.

Associative

Commutative.

Associative ( $p \wedge (p \wedge q) \Leftrightarrow p$ )

Absorption.

→

### CONVERSE INVERSE AND CONTRAPOSITIVE OF A

STATEMENT:

for a statement  $(p \rightarrow q)$ , the contrapositive of the statement is  $\neg q \rightarrow \neg p$ , the converse of the statement is  $q \rightarrow p$ .

statement:  $(p \rightarrow q)$

contrapositive:  $\neg q \rightarrow \neg p$

converse:  $q \rightarrow p$

inverse of statement:  $\neg p \rightarrow \neg q$ .

★

NOTE:

$$p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$$

$$q \rightarrow p \Leftrightarrow \neg p \rightarrow \neg q$$

Q. if the temperature is  $30^\circ\text{C}$  then i will wash my car. Write the contrapositive, converse & inverse.

Statement:  $p \rightarrow q$ .

i) contra positive: if i will not wash my car it is not  $30^\circ\text{C}$ .

ii) converse: - I will wash my car if the temp is  $30^\circ\text{C}$

iii) inverse: - if it is not  $30^\circ\text{C}$  I will not wash my car.



Q If today is labour's day then tomorrow is tuesday.

i) Contrapositive:

if today is not labour's day then tomorrow is not tuesday.

ii) inverse:

if tomorrow is not tuesday then today is not labour's day.

iii) converse:

if tomorrow is a tuesday then today is labour's day.

Q. If  $1 < 3$  and  $3+7=10$  then,  $\sin(3\pi/2)$  is equal to  $-1$ .

i) Contrapositive: if  $1$  is not lesser than  $3$  and  $3+7$  is not equal to  $10$ , then  $\sin(3\pi/2)$  is not equal to  $-1$ .

ii) inverse: if  $\sin(3\pi/2)$  is  $\neq -1$  then  $1 \nless 3$  and  $3+7 \neq 10$ .

iii) converse: if  $\sin(3\pi/2) = -1$  then  $1 < 3$  &  $3+7=10$ .

## → SIMPLIFICATION OF NETWORKS:

Between any 2 points  $T_1$  and  $T_2$ , if the input is in parallel, then the input is considered by (v) or ~~as input and if in~~ ~~par~~

in series, the input is considered by (i) and.



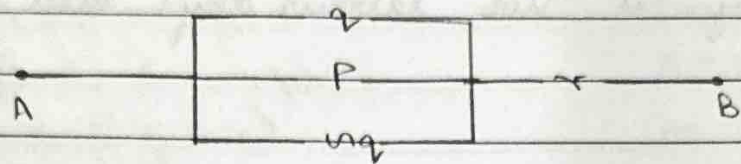
## \* NOTE:

for any simplification of network problems, it is mandatory to draw the simplified network.

→

SIMPLIFY THE FOLLOWING NETWORKS.

Q.



$$\Rightarrow (q \vee p \vee uq) \wedge r //$$

$$(q \vee p \vee uq) \wedge r$$

$$((q \vee uq) \vee p) \wedge r$$

$$(T \vee p) \wedge r$$

$$T \wedge r$$

$$r //$$

Reason

Associative.

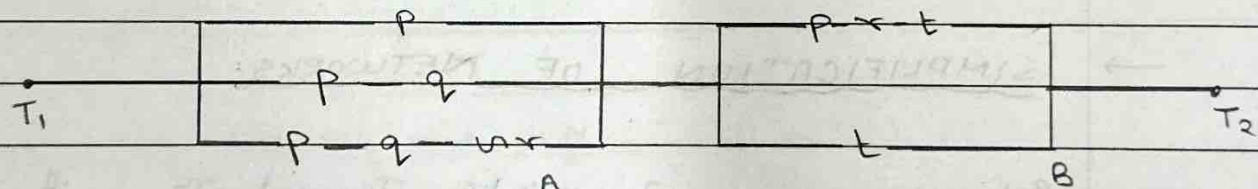
Inverse

Domination.

Identity.



Q.



$$\Rightarrow (p \vee (p \wedge q) \vee (p \wedge q \wedge ur)) \wedge ((p \wedge r \wedge t) \vee t) //$$

A:



## LOGIC IMPLICATION

- \* consider the general form of the argument

$$(P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n) \rightarrow q.$$

- where,  $n$  is a +ve integer
- $P_1, P_2, P_3 \dots P_n$  is called a premise / hypothesis of the argument
- $q$  is the conclusion of the argument
- \* The argument is called valid whenever the each of the is true and their conclusion of  $q$  is also true.
- \* The validity of the statement is represented as:

$$P \Rightarrow q$$

↳ logical implication

## THEORY OF INFERENCE

- Inference theory is concerned with inferring a conclusion from a set of premise or hypothesis by applying certain principles of reasoning called as rules of inference.
- When a conclusion is derived from a set of premises by using rules of inference, the process of such derivation is called as formal proof.
- There are 12 rules of inference.



## RULES OF INFERENCE

1) Modus ponens / Law of Detachment:

$$[p \wedge (p \rightarrow q)] \rightarrow q$$

which can also be written as

$p$

$$p \rightarrow q$$

$$\therefore q$$

examples: • Ram wins a 10 million lottery. (P)

• If Ram wins 10 million lottery then Sita will quit her job. (P)

• Therefore, Sita quits her job. (S)

2) Modus Tollens:

$$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

$$p \rightarrow q$$

$$\sim q$$

$$\therefore \sim p$$

eg: • If Yash is elected president of abc company then Rita will pledge the company. (P)

• Rita will not pledge the company. (S)

• Therefore, Yash is not elected as the president of abc company. (S)

3) Law of Syllogism:

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow [p \rightarrow r]$$

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$



eg: • If we <sup>invest</sup> are ~~not~~ in stock market, then we will get rich. (P)

• If we are rich, then we will be happy. (P)

• If we invest in stock market, then we will be happy. (S)

4) Rule of Conjunction:

$$p \wedge q \rightarrow (p \wedge q)$$

p

q

$$\therefore p \wedge q$$

eg: • It is raining

• It is windy

• Therefore, it is raining and it's windy.

5) Rule of Disjunction Syllogism:

$$(p \vee q) \wedge \sim q \rightarrow p$$

$$p \vee q$$

$$\sim q$$

$$\therefore p$$

eg: • The icecream is either vanilla flavoured or chocolate flavoured.

• The icecream is not vanilla flavoured.

• The icecream is chocolate flavoured.

6) Rule of Contradiction:

$$(\sim p \rightarrow F) \rightarrow p$$

$$\sim p \rightarrow F$$

$$\therefore p$$

eg: with truth table:

| p | $\sim p$ | F | $\sim p \rightarrow F$ | $(\sim p \rightarrow F) \rightarrow p$ |
|---|----------|---|------------------------|----------------------------------------|
| 1 | 0        | 0 | 1                      | 1                                      |
| 0 | 1        | 0 | 0                      | 1                                      |

7) Rule of Conjunctive Simplification:

$$\frac{p \wedge q}{\therefore p} \quad (\text{or}) \quad \frac{p \wedge q}{\therefore q}$$

eg: • It is raining and it is windy.  
• Therefore, it is raining.

8) Rule of Disjunctive Amplification:

$$\frac{p}{\therefore p \vee q}$$

eg: p: It is raining  
q: It is windy.

• It is raining.  
• Therefore, it is raining and it is windy.

9) Rule of Conditional Proof:

$$[(p \wedge q) \wedge (p \rightarrow (q \rightarrow r))] \rightarrow r$$

$$\frac{p \wedge q}{p \rightarrow (q \rightarrow r)} \\ \hline \therefore r$$

eg:



10) Rule of proof by case:

$$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow (p \vee q) \rightarrow r$$

$$p \rightarrow r$$

$$q \rightarrow r$$

$$\therefore (p \vee q) \rightarrow r$$

eg: • If I am feeling sad then I will go to the gym.  
• If I am feeling happy then I will go to the gym.  
• If I am feeling sad or I am feeling happy then I'll go to the gym.

11) Rule of constructive dilemma:

$$[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$$

$$p \rightarrow q$$

$$r \rightarrow s$$

$$p \vee r$$

$$\therefore q \vee s$$

eg: • If it is cloudy, I will take leave.  
• If it is hot outside, I will go to the pool.  
• Either it is cloudy or not outside.  
Therefore, either I will take leave or go to the pool.

\* Note:

Constructive dilemma and destructive dilemma are almost opposite each other.

12) Rule of Destructive Dilemma:

$$[(p \rightarrow q) \wedge (s \rightarrow r) \wedge (\sim q \vee \sim s)] \rightarrow (\sim p \vee \sim r)$$

$$p \rightarrow q$$

$$r \rightarrow s$$

$$\neg q \vee \neg s$$

$$\therefore \neg p \vee \neg s$$

eg: • if it is cloudy, I will take leave

• if it is hot outside, I will go to the pool.

• Either i will not take leave or not go to the pool.

$\therefore$  Either it is not cloudy or it is not hot outside.

→ Show & guess which rule of inference:

Q. • if you have a password, you can log into your account.

• You cannot log into your account.

• You don't have the password.

$\therefore$  Rule of Modus Tollens.

|                     |
|---------------------|
| $p \rightarrow q$   |
| $\neg q$            |
| $\therefore \neg p$ |

Q. • The icecream is either vanilla flavoured or chocolate flavoured.

• The icecream is not vanilla flavoured.

• The icecream is chocolate flavoured.

$\therefore$  Rule of disjunctive syllogism.

|                |
|----------------|
| $p \vee q$     |
| $\neg p$       |
| $\therefore q$ |

Q. • If it rains, I shall not go to college.

• If i don't go to college, i won't need to do assignments.

• If it rains, i don't need to do assignments.

$\therefore$  Rule of syllogism

|                              |
|------------------------------|
| $p \rightarrow q$            |
| $q \rightarrow r$            |
| $\therefore p \rightarrow r$ |



- Q. • If it is cloudy, i will take leave.  
 • If it is hot outside, i will go to the pool.  
 • Either it is cloudy or hot outside.  
 $\therefore$  Either i will take leave or go to the pool.  
 $\therefore$  Constructive dilemma

$$\left[ \begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ \hline p \vee r \\ \hline q \vee s \end{array} \right]$$

- Q. • If it is cloudy, I will take leave.  
 • If it is hot outside, I will go to the pool.  
 • Either it ~~is~~ is not cloudy or it is not hot outside.  
 • Either i will not take leave, or not go to pool.

$\therefore$  Destructive Dilemma

$$\left[ \begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ \hline \neg q \vee \neg s \\ \hline \therefore \neg p \vee \neg r \end{array} \right]$$

- Q. • If you send me an Email, I will finish writing the program.  
 • If you don't send me an Email, I will go to sleep.  
 • If i go to sleep then i will feel refreshed.

→ USING THE 4 PROVIDED STATEMENTS VERIFY:

- If i do not finish writing the program, then I will wake up feeling refreshed.

$p \rightarrow$  You send me an email

$q \rightarrow$  I will finish writing the program.

$r \rightarrow$  I will go to sleep.

$s \rightarrow$  I will wake up feeling refreshed.

|   | <u>Steps</u>                | <u>Reason</u>          |
|---|-----------------------------|------------------------|
| 1 | $\sim p \rightarrow r$      | premise                |
| 2 | $r \rightarrow s$           | premise                |
| 3 | $\sim p \rightarrow s$      | law of syllogism ① & ② |
| 4 | $p \rightarrow q$           | premise                |
| 5 | $\sim q \rightarrow \sim p$ | contrapositive         |
| 6 | $\sim q \rightarrow s$      | Law of syllogism ③ & ⑤ |

(another way to solve):

|   | <u>Steps</u>      | <u>Reason</u>                                                                      |
|---|-------------------|------------------------------------------------------------------------------------|
| 1 | $p \rightarrow q$ | premise                                                                            |
| 2 | $p \vee r$        | $\sim p \rightarrow r \Leftrightarrow p \vee r$                                    |
| 3 | $r \rightarrow s$ | premise.                                                                           |
| 4 | $q \vee s$        | law of constructive inference<br>$(\sim q \rightarrow s \Leftrightarrow q \vee s)$ |

Q. Show that  $(s \vee r)$  is logically implied by  $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow s)$ .

provided:  $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow s)$

given:  $(s \vee r)$

|                   |                            |
|-------------------|----------------------------|
| $p \vee q$        | } constructive<br>dilemma. |
| $p \rightarrow r$ |                            |
| $q \rightarrow r$ |                            |
| $s \vee r$        |                            |

|   | <u>Steps</u>           | <u>Reason</u>                                   |
|---|------------------------|-------------------------------------------------|
| 1 | $p \vee q$             | premise                                         |
| 2 | $\sim p \rightarrow q$ | $p \vee q \Leftrightarrow \sim p \rightarrow q$ |
| 3 | $q \rightarrow s$      | premise.                                        |
| 4 | $\sim p \rightarrow s$ | syllogism ③ & ②                                 |



Q. Check for the validity of the arguments:

$$p \rightarrow r$$

$$r \rightarrow s$$

$$t \vee \sim s$$

$$\sim t \vee u$$

$$\sim u$$

$$\therefore \sim p$$

|    | <u>STEPS</u>                | <u>REASON</u>                                   |
|----|-----------------------------|-------------------------------------------------|
| 1  | $\sim t \vee u$             | premise                                         |
| 2  | $t \rightarrow u$           | $\sim t \vee u \Leftrightarrow t \rightarrow u$ |
| 3  | $\sim s \vee t$             | premise                                         |
| 4  | $s \rightarrow t$           | $\sim s \vee t \Leftrightarrow s \rightarrow t$ |
| 5  | $s \rightarrow u$           | sylogism (4) & (3)                              |
| 6  | $p \rightarrow r$           | premise                                         |
| 7  | $r \rightarrow s$           | premise                                         |
| 8  | $p \rightarrow s$           | sylogism (6) & (7)                              |
| 9  | $p \rightarrow u$           | sylogism (8) & (5)                              |
| 10 | $\sim u \rightarrow \sim p$ | contrapositive.                                 |



## DUAL OF A STATEMENT

Let  $S$  be a statement, If  $S$  contains no logical connectives other than  $\wedge$  and  $\vee$  then, the dual of  $S$  denoted by  $S^d$  is the statement obtained from  $S$  by replacing  $\vee$  and  $\wedge$  with  $\wedge$  and  $\vee$  respectively. &  $T$  and  $F$  by  $F$  and  $T$ .

### Principle of Duality:

Let  $S$  and  $T$  be statements that contain no logical connectives other than AND and OR. IF  $S \Leftrightarrow T$  then  $S^d \Leftrightarrow T^d$ .

Q. Find the dual of a statement:

i)  $S: (p \wedge \neg q) \vee (r \wedge T_0)$

$S^d: (p \vee \neg q) \wedge (r \vee T_0)$

| $p$ | $q$ | $\neg q$ | $(p \vee \neg q)$ | $r$ | $T_0$ | $(r \vee T_0)$ |
|-----|-----|----------|-------------------|-----|-------|----------------|
| 1   | 0   | 1        | 1                 |     |       |                |
| 1   | 1   | 0        | 0                 |     |       |                |
| 0   | 0   | 1        | 0                 |     |       |                |
| 0   | 1   | 0        | 0                 |     |       |                |

Q.  $S: q \rightarrow p$   
 $(\neg q \vee p)$   
 $S^d: \neg q \wedge p$

Q.  $p \leftrightarrow q$

$S^d: (p \rightarrow q) \wedge (q \rightarrow p) \Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$

$S: (\neg p \vee q) \wedge (\neg q \vee p)$

$S^d: (\neg p \wedge q) \vee (\neg q \wedge p)$

Q.  $p \rightarrow (q \wedge r)$   
 $S: \neg p \vee (q \wedge r)$   
 $S^d: \neg p \wedge (q \vee r)$





## SUBSTITUTION RULES

1) Suppose the compound statement  $P_{(p,q,r,s)}$  is a tautology. If  $p$  is a primitive statement that appears in  $P$  and we replace each occurrence of  $p$  by the same statement  $q$ , then the resulting compound statement  $P_1$  is also a tautology.

2) Let  $P$  be a compound statement where  $p$  is an arbitrary statement that appears in  $P$ . Let  $Q$  be a statement such that  $q \Leftrightarrow p$ . Suppose that in  $P$  we replace one or more occurrences of  $p$  by  $q$ . Then, this replacement is the compound statement  $P_1$ . Under these circumstances,  $P_1 \Leftrightarrow P$ .



## QUANTIFIERS

Open Statement: A sentence can be called as an open statement if:

- \* it should contain 1 or more variables.
- \* it should not be a statement.
- \* it will become a statement when the variables are replaced with values

eg:  $x^2 > 0$

take  $x=8$

$$8^2 > 0$$

$$64 > 0 //$$



\*  $\forall$ : For all.

\*  $\exists$ : There exists atleast.

### → Universal Quantifiers:

The cases where, the variable  $x$  is considered for all the cases.

eg:  $\forall x, x \in \mathbb{N}$

### → Existential Quantifiers:

If for a few of the cases, if the variable  $x$  is considered then it is ~~referred~~ referred as an existential quantifier.

eg:  $\exists x, x \in y = mx + c$ .

\* Suppose  $P(x)$  indicates a predicate where  $x$  must take an electronic force/course. And  $Q(x)$  also indicates a predicate where  $x$  is an electrical student.

Q. Represent the given statements in terms of quantifiers.

i) \* Every electronics student must take an electronic course.

\* Everybody must take an electronics course or be an electrical student.

$$\forall x [q(x) \rightarrow p(x)].$$

$$\forall x [q(x) \vee p(x)].$$



2) \* Suppose  $P(x)$  indicates a predicate where  $x$  is a square and  $Q(x)$  indicates a predicate where  $x$  is a rectangle. With the help of the quantifiers represent the following:

i) All squares cannot be a rectangle.

ii) Atleast one rectangle is a square.

Sol. i)  $\forall x [P(x) \rightarrow \neg Q(x)]$

ii)  $\exists x [Q(x) \rightarrow P(x)]$ .

### → The rule of universal specification

\* If an open statement becomes true for all replacements by the members in the given universe, then that open statement is true for each specific individual member in the universe.

\* Symbolically if  $P(x)$  is an open statement for a given universe,  $\forall x P(x)$  is true then  $P(a)$  is true for each 'a' in the universe.

\* eg: Q. For the universe of all the people consider the open statements:-

•  $m(x)$ ,  $x$  is a mathematics professor.

•  $c(x)$ ,  $x$  has studied calculus.

Consider the following statements:

\* All mathematics teachers have studied calc.

\* Dana is a mathematics professor.

Sol:  $\forall x [m(x) \rightarrow c(x)]$   $P(x) \rightarrow \text{True}$

$m(a)$

$P(a) \rightarrow \text{True}$ .

$\therefore c(a)$



- Q.  $P(t)$  :  $t$  has two sides of equal length.  
 $Q(t)$  :  $t$  is an isosceles triangle.  
 $R(t)$  :  $t$  is two angles of equal measurements.

- in a triangle  $(x, y, z)$  there is no pair of angles of equal measurement.
- if a triangle has two sides of equal length then it is isosceles.
- If a triangle is isosceles then it has two angles of equal measurement.

**\*NOTE** : it is on the set of all triangles.

Sol:  $\sim \forall x (a)$

$$\forall t [P(t) \rightarrow Q(t)].$$

$$\forall t [Q(t) \rightarrow R(t)].$$

### → RULE OF UNIVERSAL GENERALISATION.

- \* If an open statement  $p(x)$  is proved to be true, when  $x$  is replaced by any arbitrarily chosen element  $c$  from the universe then, the universally quantified statement for all  $x$ ,  $p(x)$  is true.

$$\Rightarrow \forall x p(x) \text{ is true.}$$

- \* The rule extends beyond a single variable, if we have an open statement  $q(x, y)$  that is proved to be true, when  $x$  and  $y$  are replaced by arbitrarily chosen elements from the same universe (or) their own respective universe then, the universally quantified statement

$$\Rightarrow \forall x \forall y q(x, y) \text{ (or) } \forall x, y q(x, y) \text{ is true.}$$



**\* NOTE:** let  $p(x)$ ,  $q(x)$  and  $r(x)$  be open statements that are defined for a given universe, prove the following:

$$\begin{aligned} &* \forall x [p(x) \rightarrow q(x)] \\ &\quad \forall x [q(x) \rightarrow r(x)] \\ &\therefore \forall x [p(x) \rightarrow r(x)]. \end{aligned}$$

| STEPS                                  | REASON                                    |
|----------------------------------------|-------------------------------------------|
| 1) $\forall x [p(x) \rightarrow q(x)]$ | Premise (RUS).                            |
| 2) $p(a) \rightarrow q(a)$             | Rule of Universal specification on step ① |
| 3) $\forall x [q(x) \rightarrow r(x)]$ | Premise                                   |
| 4) $q(a) \rightarrow r(a)$             | RUS on step ③.                            |
| 5) $p(a) \rightarrow r(a)$             | ② & ④: Law of syllogism.                  |
| 6) $\forall x [p(x) \rightarrow r(x)]$ | ⑤: Rule of universal generalization (RUG) |

**\* Note:**

- for the problems involving quantifiers, the laws of logic and rules of inferences cannot be applied directly.
- The rules of inference and laws of logic can only be applied for individual entities.

### → RULES FOR NEGATING STATEMENTS WITH 1 QUANTIFIER:

$$1) \neg [\forall x p(x)] \Leftrightarrow \exists x [\neg p(x)].$$

$$2) \neg [\exists x p(x)] \Leftrightarrow \forall x [\neg p(x)].$$

$$3) [\forall x \neg p(x)] \Leftrightarrow \exists x \neg \neg p(x) \Leftrightarrow \exists x p(x).$$

$$4) [\exists x \neg p(x)] \Leftrightarrow \forall x \neg \neg p(x) \Leftrightarrow \forall x p(x).$$



→ VERIFY WHETHER THE STATEMENTS ARE TRUE OR FALSE.

- Q.  $p(n) : n \geq 0.$   
 $q(n) : n^2 \geq 0.$   
 $r(n) : n^2 - 3n - 4 = 0.$   
 $s(n) : n^2 - 3 > 0.$

F statement 1:  $\exists n [p(n) \wedge r(n)].$

statement 2:  $\forall n [p(n) \rightarrow q(n)]$

statement 3:  $\forall n [q(n) \rightarrow s(n)]$

statement 4:  $\exists n [r(n) \vee s(n)]$

statement 5:  $\forall n [r(n) \rightarrow p(n)].$

statement 6:  $\exists n [p(n) \rightarrow q(n)].$

}  $n$  is on the set of  
integer universe.

→ PROOF OF THEOREMS

Let  $n$  be an integer,  $n$  is even if it is divisible by two. i.e. there exists an integer  $r$  such that,  $n = 2r$ . If  $n$  is odd, there exists an integer  $s$  where,  $n = 2s + 1$ .

Theorem 1: For all the integers,  $k$  and  $L$ . if  $k$  and  $L$  are both odd then,  $k+L$  is even.

Three proofs of this theorem are:

- direct proof
- contrapositive proof
- 

→



- Direct Proof :

$$k : 2s+1, \quad l = 2a+1.$$

$$k+l = (2s+1) + (2a+1).$$

$$\Rightarrow 2s+2a+2$$

$$\Rightarrow 2[(s+a+1)].$$

$$\text{considering } s+a+1 = r$$

$$k+l = 2r$$

$\therefore k+l$  is even.

- THEOREM 2 :  $\forall k$  and  $l$ , if both are odd then their product  $k \times l$  is also odd.

$$k = 2a+1$$

$$l = 2b+1$$

$$k \times l = (2a+1)(2b+1).$$

$$k \times l = 2(2ab + a + b) + 1.$$

$$\text{considering, } 2ab + a + b = c$$

$$k \times l = 2c + 1$$

$$= \text{odd} //$$





- THEOREM 3: IF  $m$  is an even number, then  $m+1$  is odd.

• direct proof:  $m = 2a$ .

$$m+1 = 2a+1$$

$$m = (2a+1)+1$$

$$= 2a+2$$

$$= \text{odd}.$$

Hence,  $m+1$  is odd.

• proof by contrapositive:  $p$ :  $m$  is even

$q$ :  $m+1$  is odd.

assume  $\neg q$ :  $m+1$  is even.

$$m+1 = 2a$$

$$m = 2a-1$$

$$= (2a-2)+1$$

$$= 2(a-1)+1$$

considering  $a-1 = n$ .

$$m = 2n+1$$

Hence, it is odd.

★ NOTE: in the proof by contradiction we assume  $p(m)$  and  $\neg q(m)$ . The result derived at the end should be a false statement.

• direct proof by contradiction:

assume,  $m = 2a$

$$m+1 = 2a$$

$$m = 2a-1$$

$$= 2a-2+1$$

$$= 2(a-1)+1$$

$$= 2a+1$$

Hence,  $m = \text{odd}$